Artificial Intelligence Engineering Department
Bahçeşehir University

QUIZ #2 for Section 3

This is a 30-minute quiz

Student ID:	Student Full Name:

Write a pandas program that creates a DataFrame named sales with columns **'Region': ['North', 'South', 'North', 'East', 'South', 'East', 'North', 'South'], 'Q1': [120, 85, 200, 95, 60, 175, 110, 90], 'Q2': [135, 90, 190, 80, 75, 160, 130, 95], and 'Q3': [150, 70, 210, 100, 85, 180, 125, 80]**. Call **describe()** on the numeric columns and separately print the **idxmax()** and **idxmin()** of column '**Q1**' to identify the rows with the highest and lowest first-quarter sales. Compute the **cumsum()** of **'Q1'** and add it as a new column named '**Q1_cumulative**'. Then compute and print the full Pearson correlation matrix for the three quarterly columns using **.corr()**, and compute the covariance between **'Q1'** and **'Q3'** using **.cov()**. Finally, on the **'Region'** column, call **unique()** to list distinct regions, **value_counts()** to show how many rows belong to each region, and **isin(['North', 'East'])** to filter and print only the rows from those two regions.